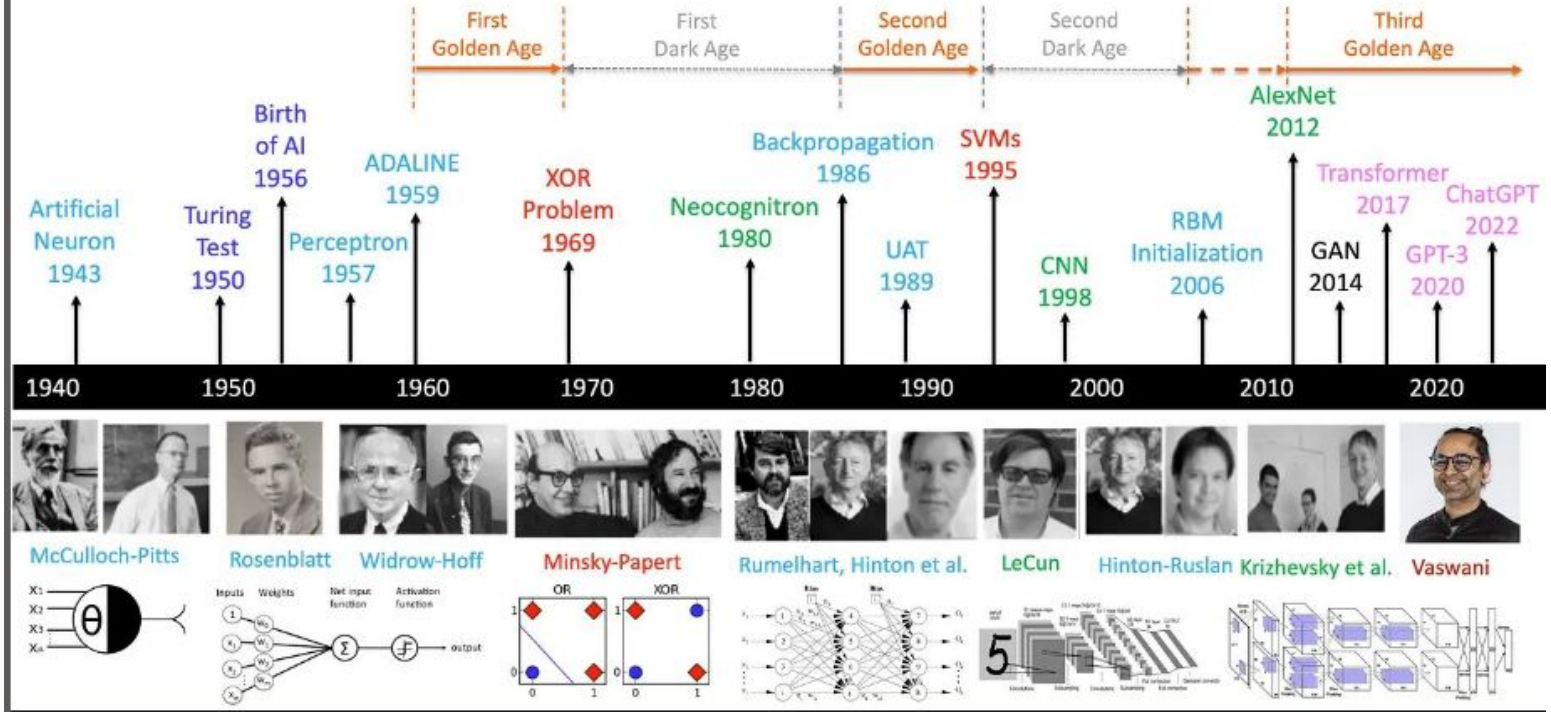
História da Deep Learning Leitura 1 (em construção)

Paulo Vieira

Agenda

A Brief History of AI with Deep Learning



History, step by step

history, highlighting its origins, evolution and the innovators who shaped it — from its early beginnings to today's transformative advancements.

- 1943: **McCulloch** & **Pitts** create the first artificial neuron.
- 1950: **Alan Turing** introduces the Turing Test, forever changing the way we view intelligence.
- 1956: **John McCarthy** coins the term “Artificial Intelligence,” marking the official birth of the field.
- 1957: **Frank Rosenblatt** invents the Perceptron, one of the first neural networks.
- 1959: **Bernard Widrow** and **Ted Hoff** create ADALINE, a model that would shape neural networks.
- 1969: **Marvin Minsky e Seymour Papert** demonstram que redes neuronais de camada única (Perceptrons) não conseguem resolver problemas não linearmente separáveis, como o XOR, contribuindo para o início do primeiro “AI Winter”.
- 1980: **Kunihiko Fukushima** introduces Neocognitron, laying the groundwork for deep learning.
- 1986: **Geoffrey Hinton** and **David Rumelhart** introduce backpropagation, making neural networks viable again.
- 1989: **Judea Pearl** advances UAT (Understanding and Reasoning), building a foundation for AI's logical abilities.
- 1995: **Vladimir Vapnik** and **Corinna Cortes** develop Support Vector Machines (SVMs), a breakthrough in machine learning.
- 1998: **Yann LeCun** popularizes Convolutional Neural Networks (CNNs), revolutionizing image recognition.
- 2006: **Geoffrey Hinton** and **Ruslan Salakhutdinov** introduce deep belief networks, reigniting interest in deep learning.
- 2012: **Alex Krizhevsky** and **Geoffrey Hinton** launch AlexNet, sparking the modern AI revolution in deep learning.
- 2014: **Ian Goodfellow** introduces Generative Adversarial Networks (GANs), opening new doors for AI creativity.
- 2017: **Ashish Vaswani** and team introduce Transformers, redefining natural language processing (NLP).
- 2020: OpenAI unveils GPT-3, setting a new standard for language models and AI’s capabilities.
- 2022: OpenAI releases ChatGPT, democratizing conversational AI and bringing it to the masses.

So, it took decades of trial, error and innovation to get us to where we are today, with two hype cycles along the way. It's an exciting and transformative time in human history and we're very fortunate to witness it unfold right in front of us!

Darmouth conference 1956

1. The Birth of Artificial Intelligence (1956)

Marvin Minsky, Claude Shannon, Ray Solomonoff and other scientists at the Dartmouth Summer Research Project on Artificial Intelligence (Photo: Margaret Minsky) Source

1.1 The Evolution of AI: From Rule-Based Systems to Deep Learning

DL ⊂ NN ⊂ ML ⊂ AI

Artificial Intelligence (AI)

- AI refers to a set of technique that enable computers **to mimic human behavior**.

Machine Learning (ML)

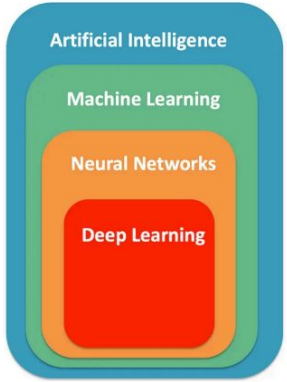
- ML is a subfield of AI , which enables machines **to learn and make prediction based on data**.

Neural Networks (NN)

- NN are a subfield of ML that **use artificial neural networks** to extract patterns from data, inspired by the human brain.

Deep Learning (DL)

- DL is a subfield of NN that **utilizes multi-layered neural networks** to achieve high performance on complex tasks.



2.1 Muclloch-Pitts Neuron (1943)

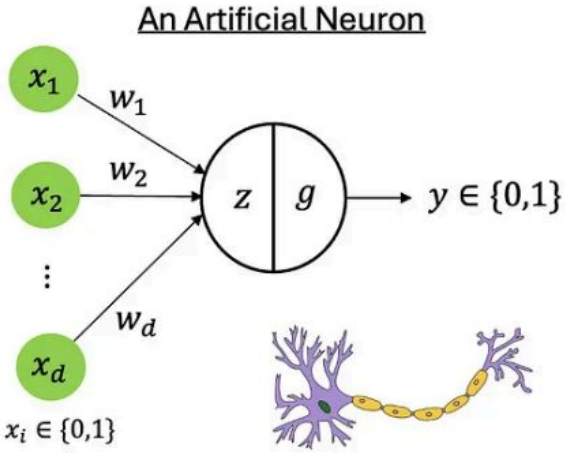
Both inputs and output are binary

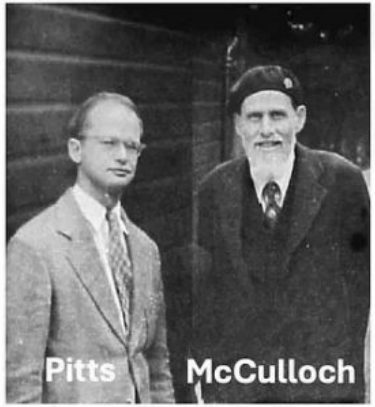
- $x_i \in \{0,1\}$ and $y \in \{0,1\}$

$$z(x_1, x_2, , \dots, x_d) = z(\mathbf{x}) = \sum_{j=1}^{d} w_j \cdot x_j$$

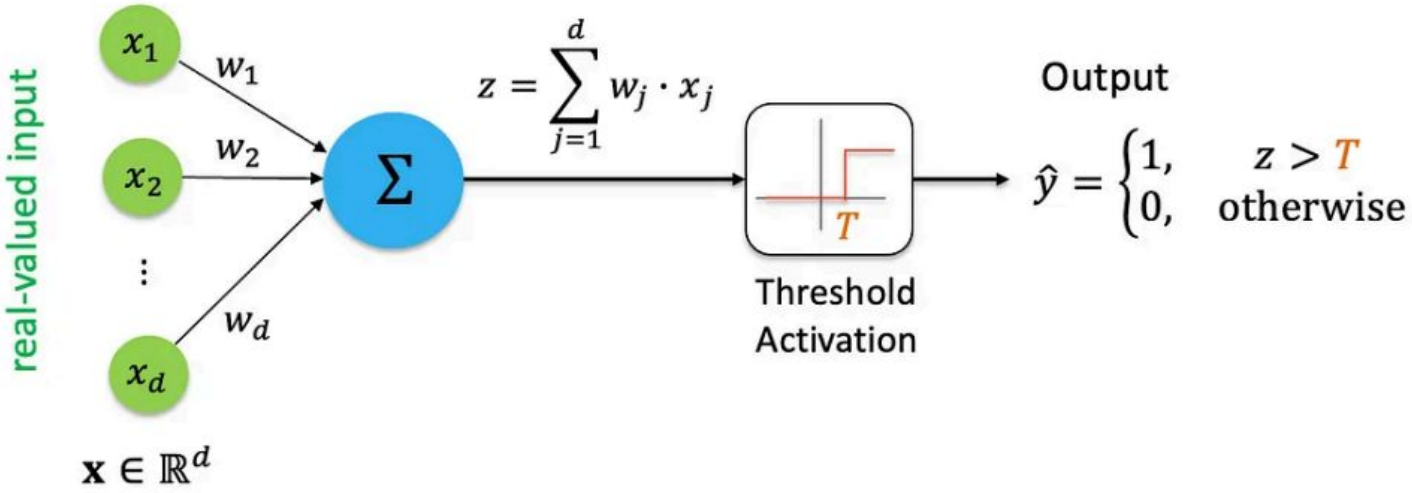
$$y = g\big(z(\mathbf{x})\big) = \begin{cases} 1 & \text{if } z(\mathbf{x}) \geq T \\ 0 & \text{if } z(\mathbf{x}) < T \end{cases}$$

Threshold Activation function





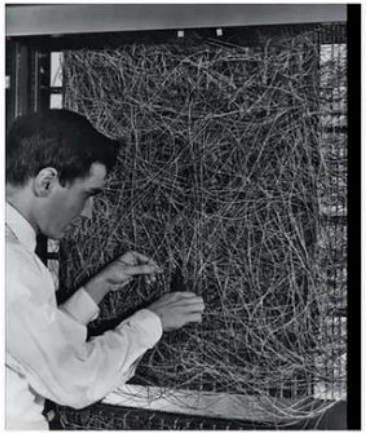
2.2 Rosenblatt's Perceptron Model (1957)



Frank Rosenblatt

2.2 Rosenblatt's Perceptron Model (1957)

$$\mathcal{L}(\mathbf{w}) = -\sum_{i \in \mathcal{M}} \mathbf{w}^{\mathrm{T}} \mathbf{x}_i y_i$$



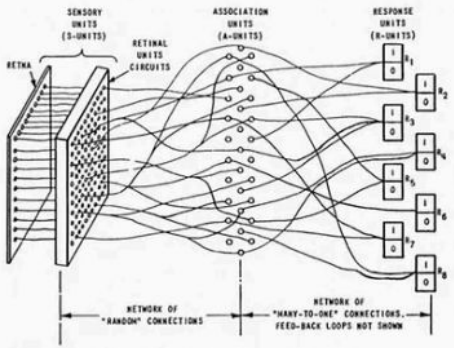
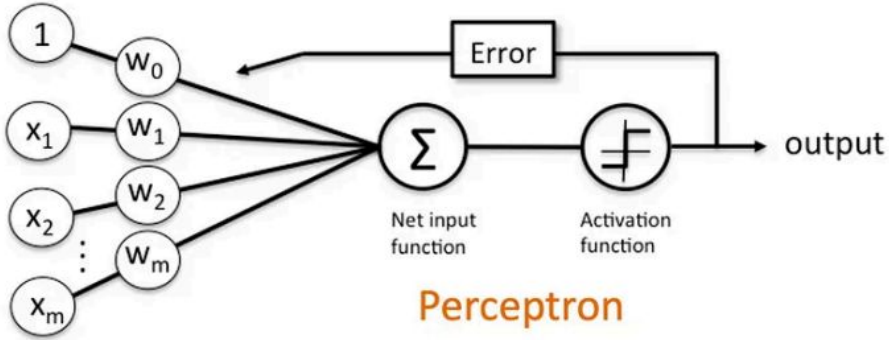


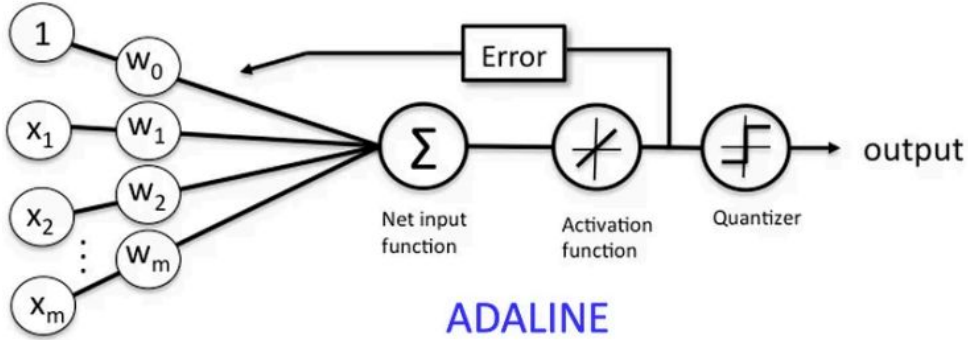
Figure 1 ORGANIZATION OF THE MARK I PERCEPTRON

The Mark I Perceptron was a machine that implemented the perceptron algorithm for image recognition.

2.3 ADALINE (1959)



Bernard Widrow



Marcian Hoff

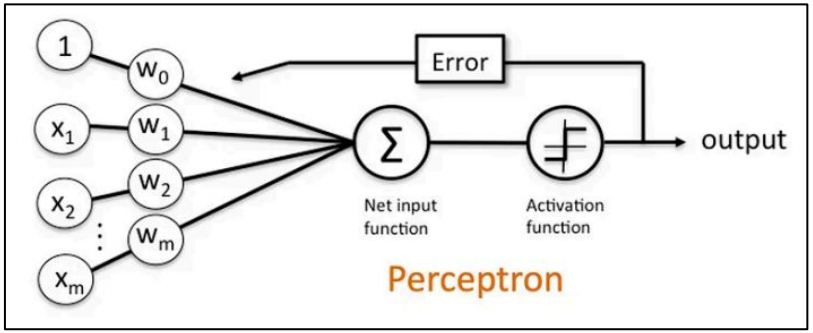
2.3 ADALINE (1959)

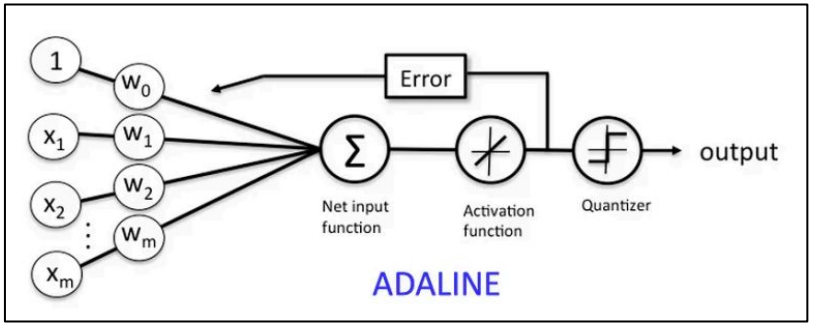
Key features of ADALINE include:

Linear Activation Function: Unlike the Perceptron's step function, ADALINE uses a linear activation function, making it suitable for regression tasks and continuous outputs.

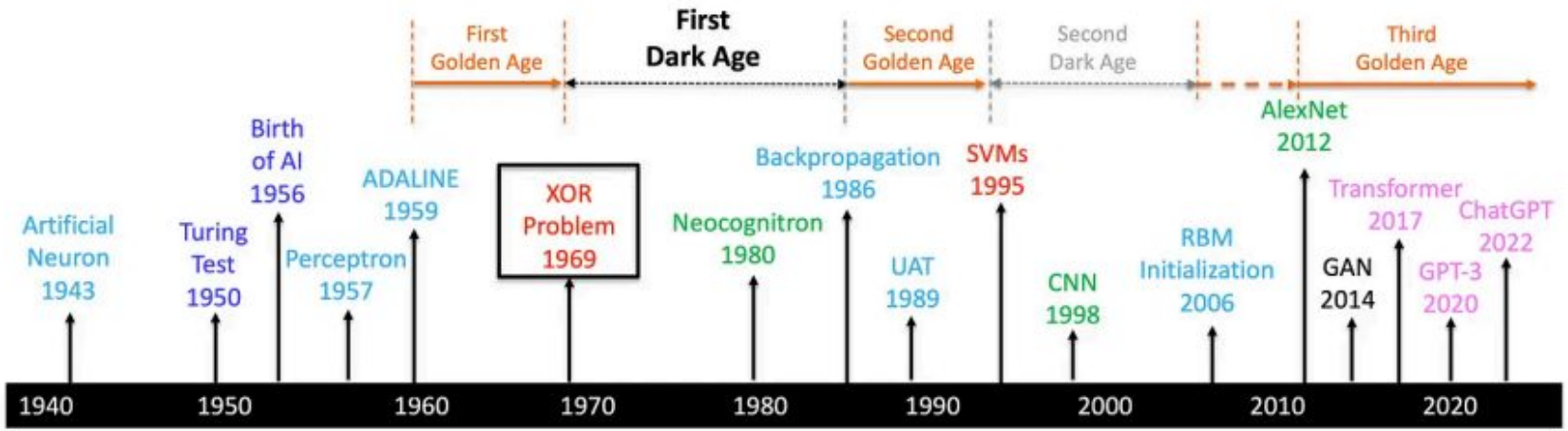
Least Mean Squares (LMS) Algorithm: ADALINE employs the LMS algorithm, which minimizes the mean squared error between predicted and actual outputs, providing a more efficient and stable learning process.

Adaptive Weights: The LMS algorithm adjusts the weights adaptively based on the error in the output, enabling ADALINE to learn and converge effectively, even in the presence of noise.





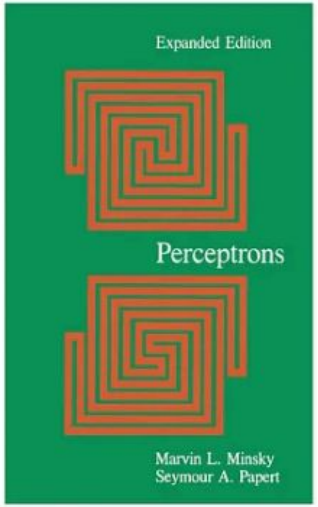
ADALINE → First Golden Age

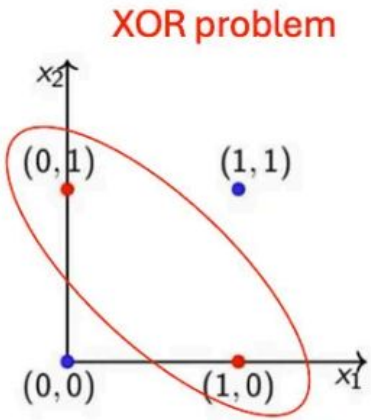


First Dark Age of Neural Network.

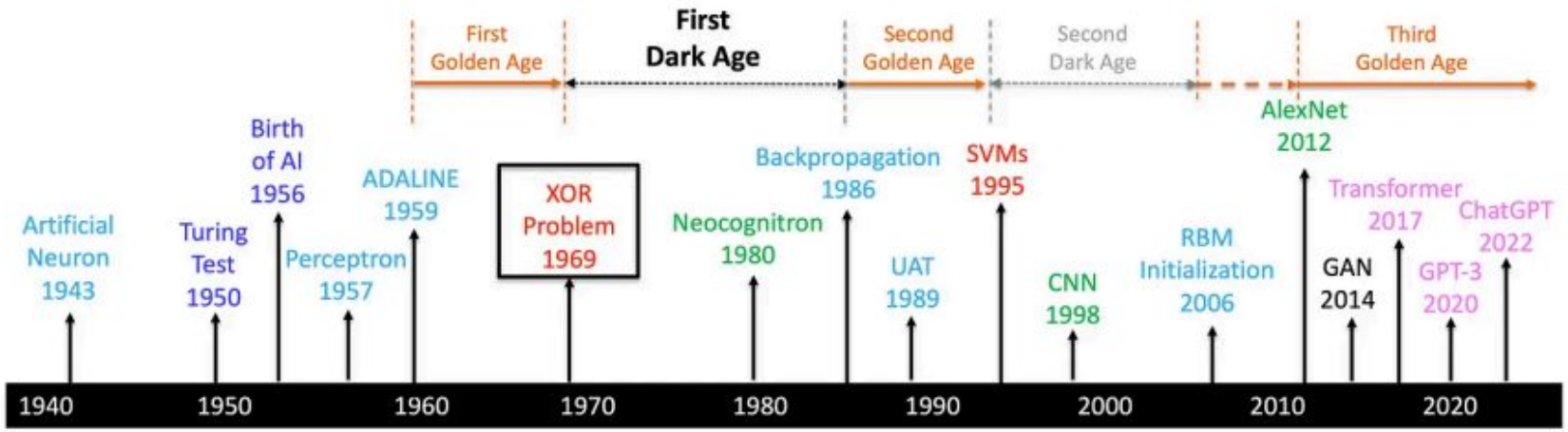
2.4 The XOR Problem (1969)

Minsky and **Papert**: Perceptrons: An introduction to computational geometry. MIT Press, 1969.



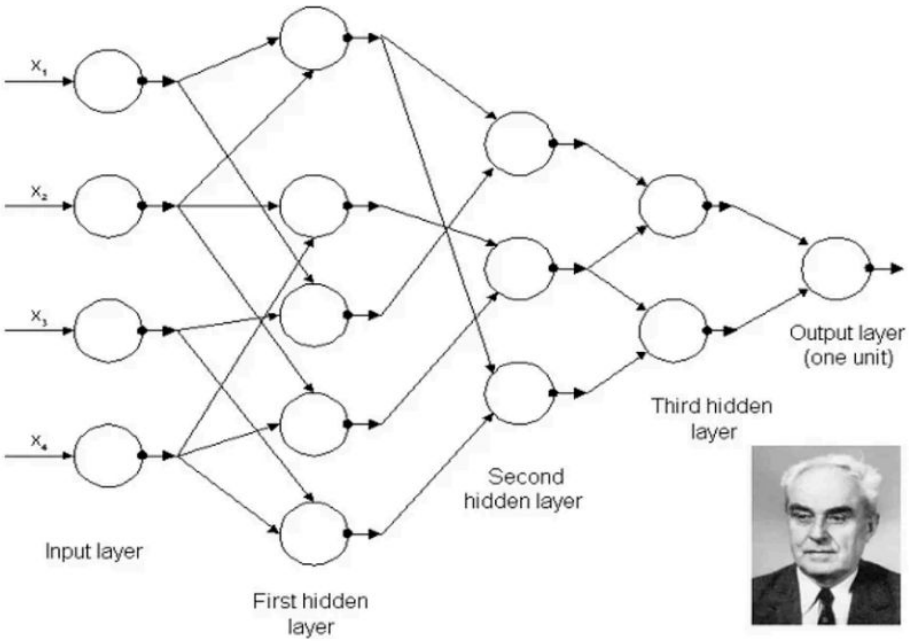


XOR problem → First Dark Age



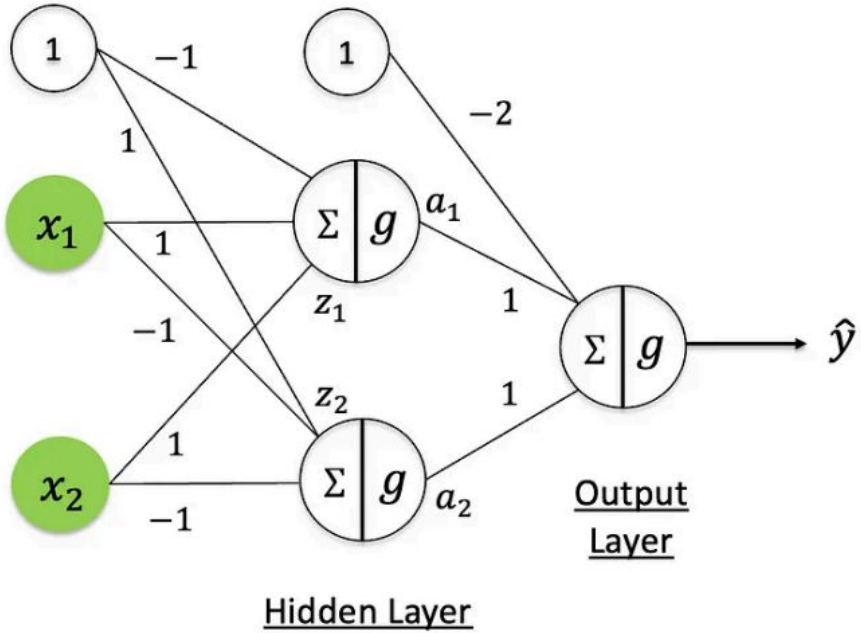
First Dark Age of Neural Network.

3. The Multilayer Perceptron (1960s)



Soviet scientists A. G. Ivakhnenko and V. Lapa made significant contributions to the development of the MLP

3.1 Hidden Layers



XOR

(x_1, x_2)	(z_1, z_2)	(a_1, a_2)	$\hat{y}$
$(0,0)$	$(-1,1)$	$(0,1)$	0
$(0,1)$	$(0,0)$	$(1,1)$	1
$(1,0)$	$(0,0)$	$(1,1)$	1
$(1,1)$	$(1,-1)$	$(1,0)$	0

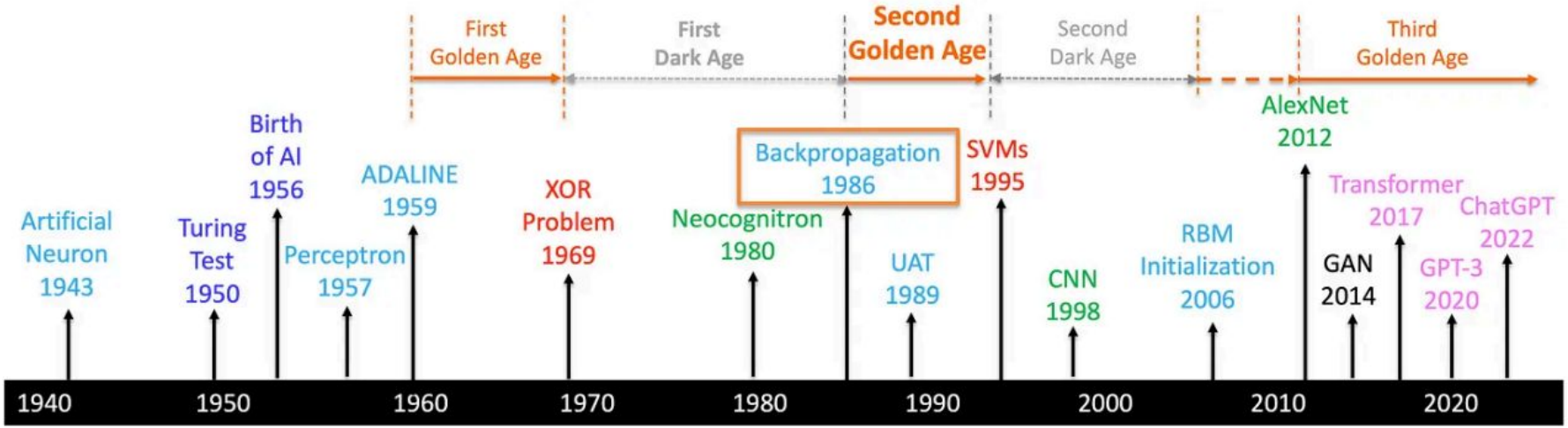
$$a = g(z) = u(z) = \begin{cases} 1, & z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Activation function is the unit step function $u(z)$

3.2 Historical Context and Challenges of MLPs

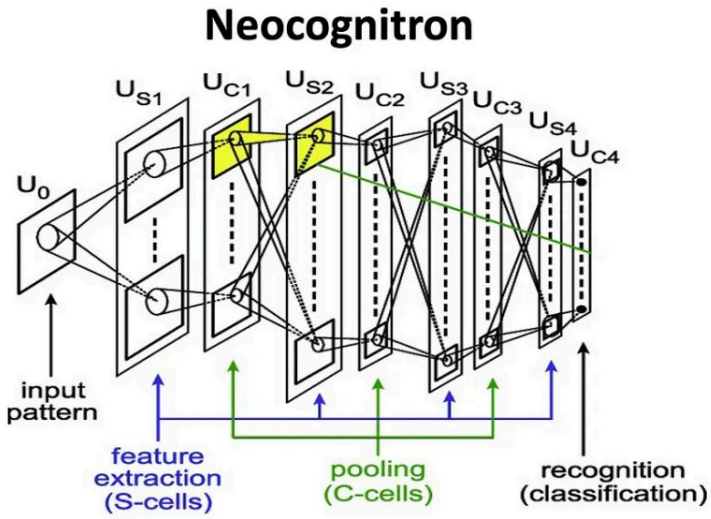
Lack of Training Algorithms: Early MLP models lacked efficient training algorithms that could effectively adjust the weights of the network. The absence of backpropagation made it difficult to train deep networks with multiple layers.

Computational Limitations: The computational power available at the time was insufficient to handle the complex calculations required for training deep neural networks. This limitation slowed the progress of MLP research and development.



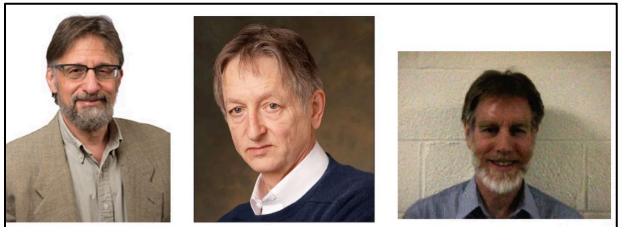
5. Convolutional Neural Networks (1980s - 2010s)

Kenji Fukushima, who proposed the Neocognitron, a hierarchical neural network that mimicked the structure of the human visual cortex. This pioneering work laid the foundation for the development of CNNs



Kenji Fukushima

4. Backpropagation (1970s-1980s)



David Rumelhart Geoffrey Hinton Ronald Williams

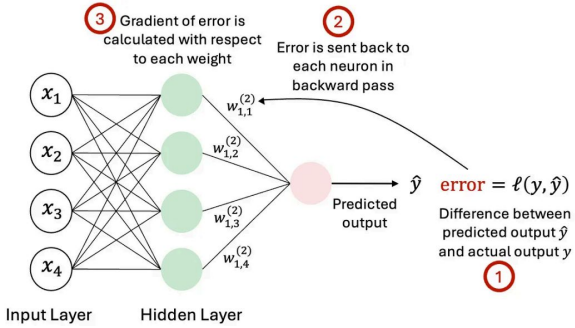
4.1 Early Developments (1970s)

- Seppo Linnainmaa (1970): Introduced the concept of automatic differentiation, which is a key component of the backpropagation algorithm.

- Paul Werbos (1974): Proposed using the chain rule of calculus to compute the gradient of the error function with respect to the network's weights, enabling the training of multilayer neural networks.

4.2 Refinement and Popularization (1980s)

David Rumelhart, Geoffrey Hinton, and Ronald Williams (1986): Presented backpropagation as a practical and efficient method for training deep neural networks, demonstrating its application to various problems.



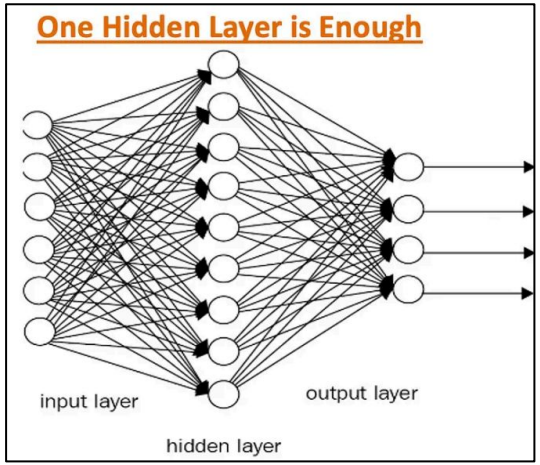
4.3 Key Features of Backpropagation:

Gradient Descent: Backpropagation is used in conjunction with gradient descent to minimize the error function. The algorithm computes the gradient of the error with respect to each weight in the network, allowing the weights to be updated iteratively to reduce the error.

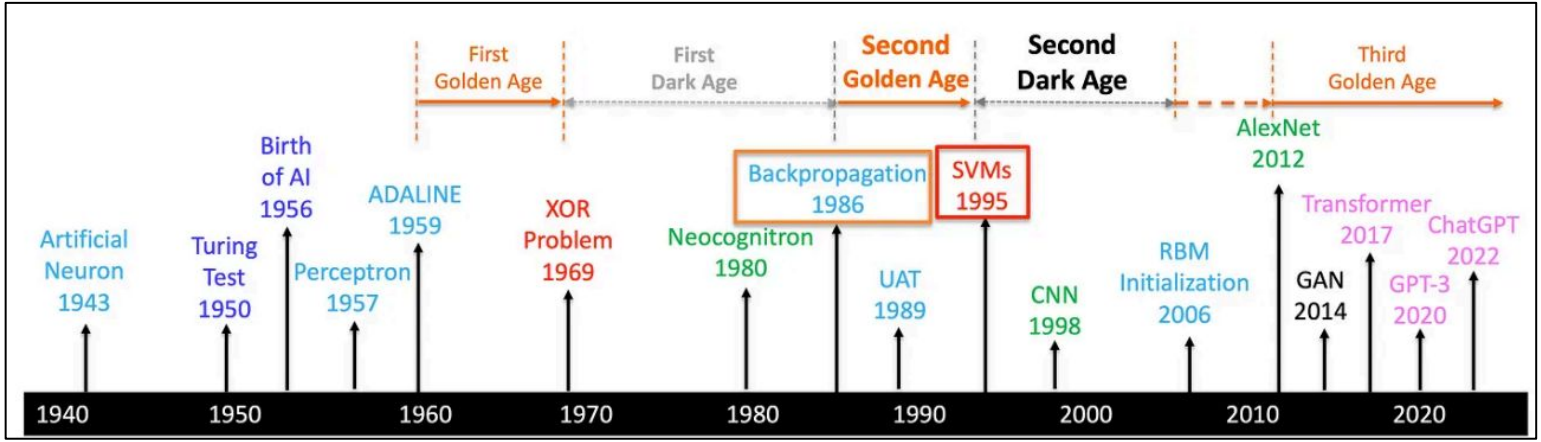
Chain Rule: The core of the backpropagation algorithm is the application of the chain rule of calculus. This rule allows the gradient of the error to be decomposed into a series of partial derivatives, which can be computed efficiently through a backward pass through the network.

Layered Computation: Backpropagation operates in a layer-by-layer manner, starting from the output layer and working backward to the input layer. This layered computation ensures that the gradients are propagated correctly through the network, enabling the training of deep architectures.

4.5 Universal Approximation Theorem (1989)



4.6 Second Golden Age (Late 1980s — Early 1990s)



4.5 Universal Approximation Theorem (1989)

4.7 Second Dark Ages (Early 1990s — Early 2000s)

The rise of Support Vector Machines (SVMs), which offered a mathematically elegant approach to classification and regression tasks.

Computational limitations, as training deep neural networks was still time-consuming and hardware-intensive.

Overfitting and generalization issues, where early neural networks performed well on training data but poorly on unseen data, making them less reliable for practical applications.

4.8 Resurgence as Deep Learning (Late 2000s — Present):

The field of neural networks experienced a resurgence in the late 2000s and early 2010s, driven by advancements in:

Deep learning architectures (CNNs, RNNs, Transformers,

Diffusion Models) Hardware (GPUs, TPUs, LPUs)

Large-scale datasets (ImageNet, COCO, OpenWebText, WikiText, etc.)

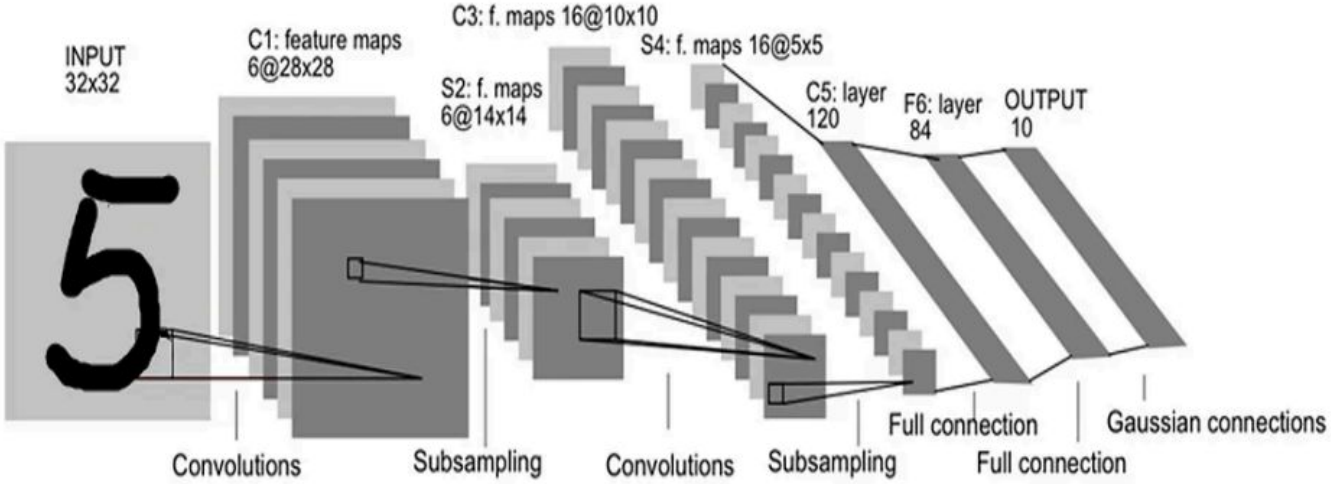
Training algorithms (SGD, Adam, dropout)

These advancements led to significant breakthroughs in computer vision, natural language processing, speech recognition, and reinforcement learning.
The Universal Approximation Theorem, combined with practical advancements, has paved the way for the widespread adoption and success of deep learning techniques.

5. Convolutional Neural Networks (1980s - 2010s)

5.1 Early Developments (1989–1998)

In the late 1980s and early 1990s, Yann LeCun and his team further developed CNNs, introducing the LeNet-5 architecture, which was specifically designed for handwritten digit recognition.



LeNet-5

Yann LeCun

5.2 Key Components of CNNs

CNNs are constructed by three key components:

1. Convolutional Layers: These layers automatically learn spatial hierarchies of features from input images by applying a set of learnable filters.

2. Pooling Layers: Pooling layers reduce the spatial dimensions of the input, enhancing robustness to variations and decreasing computational load.

3. Fully Connected Layers: Following convolutional and pooling layers, fully connected layers are used for classification tasks, integrating features learned from previous layers.

5.3 Key Features of CNNs

Local Receptive Fields: CNNs use local receptive fields to capture local patterns in the input data, making them highly effective for image and visual tasks.

Shared Weights: The use of shared weights in convolutional layers reduces the number of parameters in the network, making it more efficient and easier to train.

Translation Invariance: Pooling layers introduce translation invariance, allowing the network to recognize patterns regardless of their position in the input image.

Boltzman machines

Definition

Boltzmann Machines are **probabilistic generative neural networks** that learn the **probability distribution of data**.

Introduced by

Geoffrey Hinton & Terry Sejnowski (1980s)

Key Idea

The network learns by **minimising an energy function**, where lower energy states correspond to **more probable data configurations**.

$$E(v, h) = -\sum w_{ij} v_i h_j$$

Structure

```
Visible layer       Hidden layer

v1   v2   v3   v4
 |    |    |    |
 |    |    |    |
h1   h2   h3   h4
```

Limitation

Full Boltzmann Machines are **computationally expensive to train**.

Solution

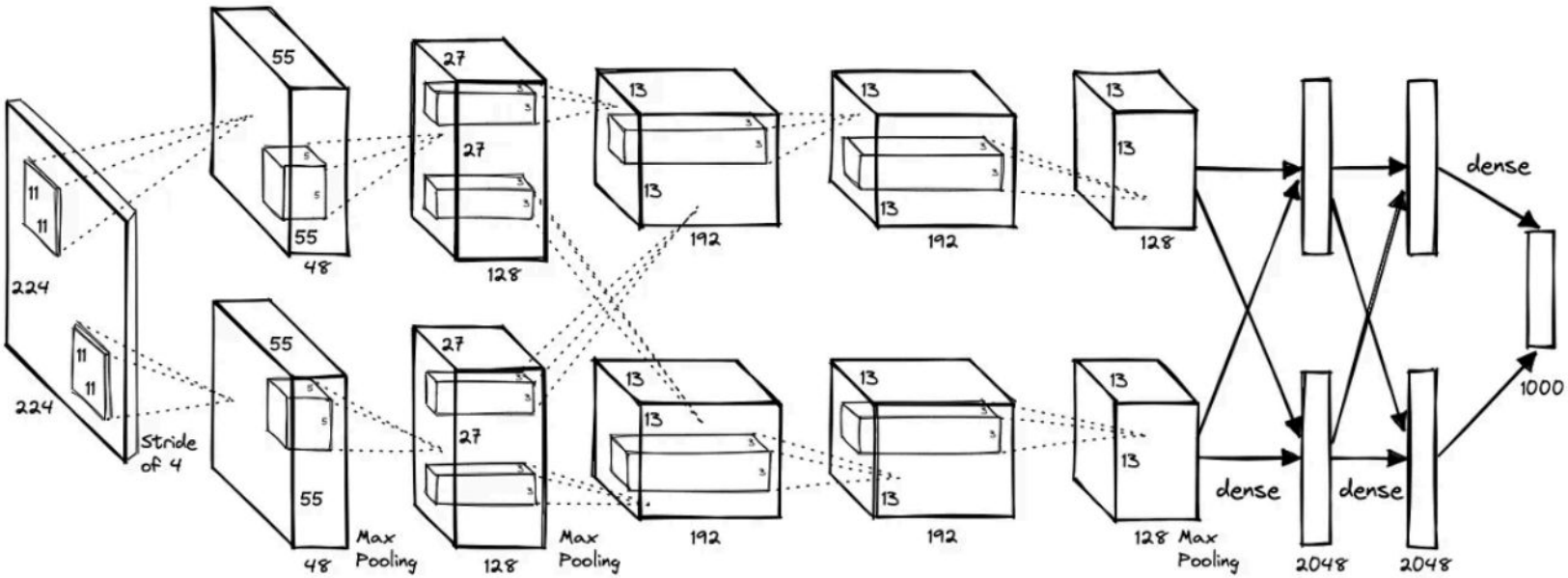
Restricted Boltzmann Machines (RBM) remove connections within the same layer, making training easier.

Importance

RBMs were used in **Deep Belief Networks (2006)** and helped revive interest in **Deep Learning**.

5.4 The Rise of CNNs: AlexNet's Impact (2012)

In 2012, a major milestone was reached in the development of CNNs, as AlexNet emerged victorious in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), achieving a significant margin of victory and marking a significant breakthrough in image classification.



Architecture of the AlexNet (2012)

5.4 The Rise of CNNs: AlexNet’s Impact (2012)

The ILSVRC is an annual image recognition benchmark that evaluates algorithms on a dataset of over 10 million annotated images, categorized into 1000 classes.

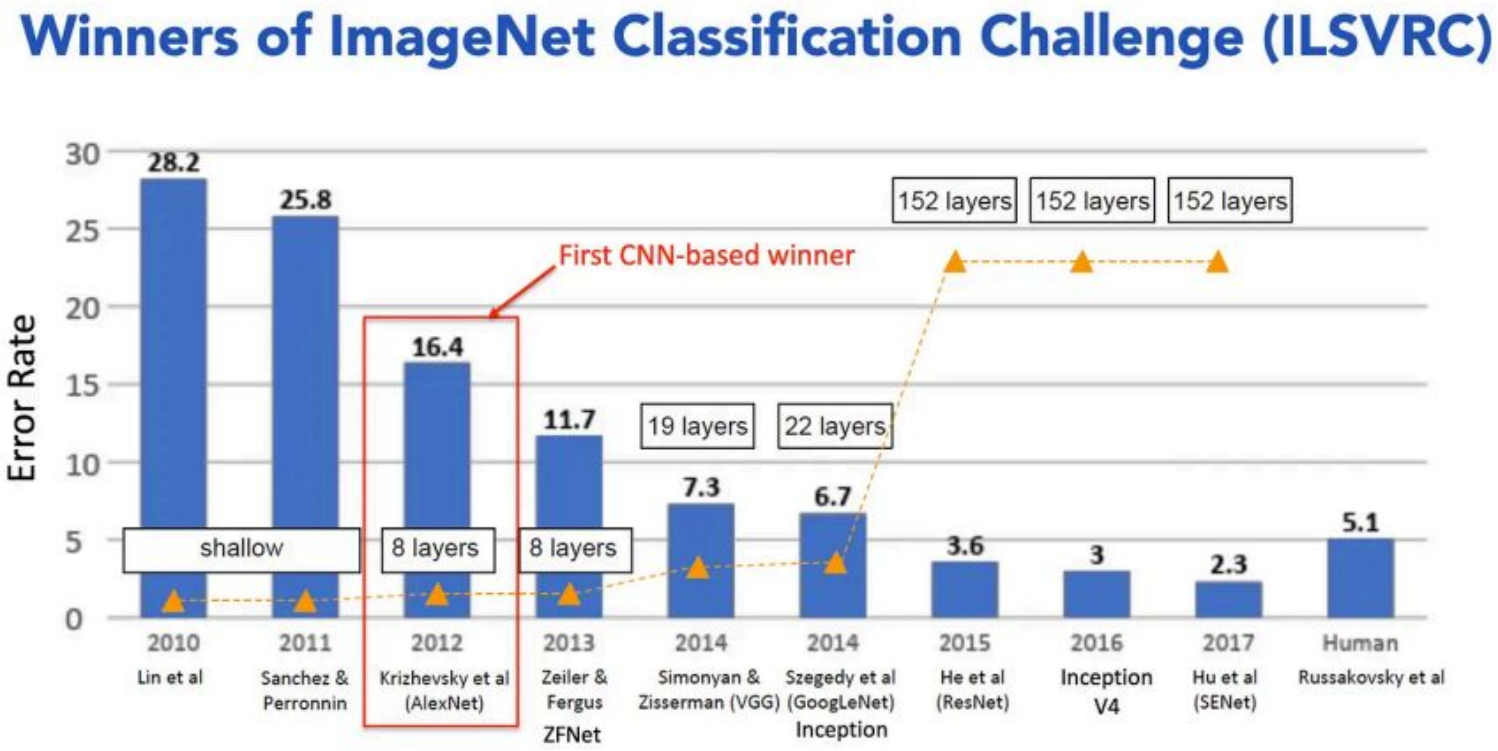
AlexNet’s innovations included:
1. ReLU Activation Functions: Introduced to overcome issues with traditional activation functions, ReLU enabled faster training and improved performance.

2. Dropout Regularization: This technique reduced overfitting by randomly dropping units during training.

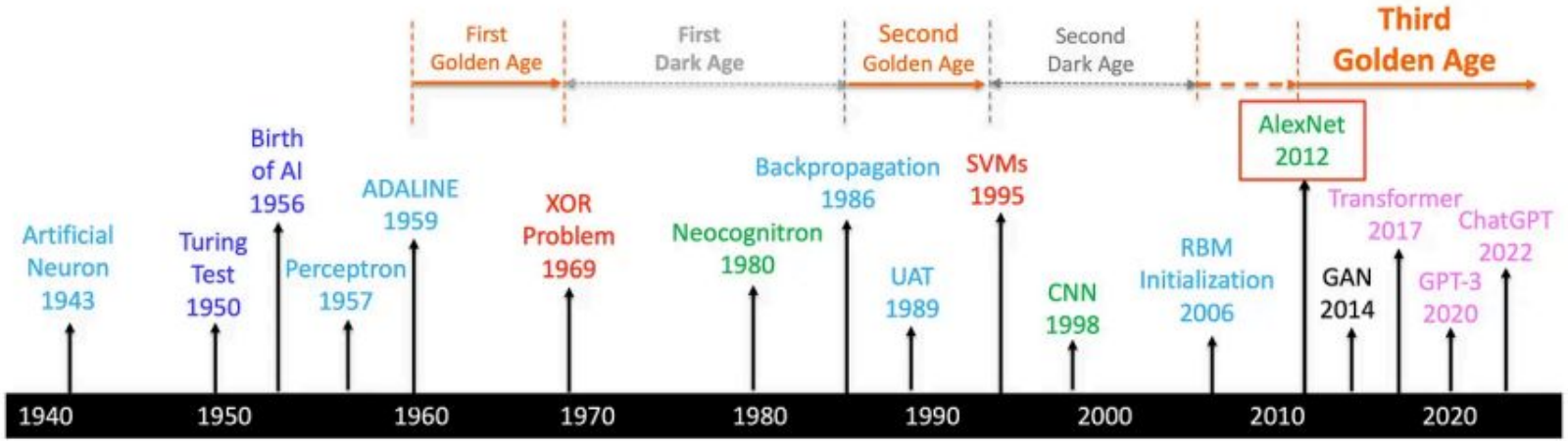
3. Data Augmentation: Enhancements to the training dataset improved generalization by artificially increasing the diversity of the training data.

AlexNet’s success marked a turning point in CNN development, paving the way for further advancements in image classification and object detection.

ILSVRC


Winners of ImageNet Classification Challenge (ILSVRC)
Error Rate
30
25
20
15
10
5
0
28.2
25.8
16.4
11.7
7.3
6.7
3.6
3
2.3
5.1
shallow
8 layers
8 layers
19 layers
22 layers
152 layers
152 layers
152 layers
First CNN-based winner
2010
Lin et al
2011
Sanchez & Perronnin
2012
Krizhevsky et al (AlexNet)
2013
Zeiler & Fergus
ZFNet
2014
Simonyan & Zisserman (VGG)
2014
Szegedy et al (GoogLeNet)
Inception
2015
He et al (ResNet)
2016
Inception V4
2017
Hu et al (SENet)
Human
Russakovsky et al

AlexNet Unleashes the Third Golden Age of Neural Networks



5.5 Subsequent Architectures

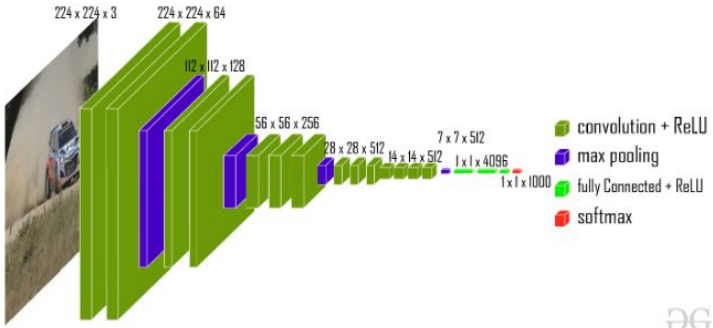
Following AlexNet, several influential architectures emerged:

VGGNet (2014): Developed by the Visual Geometry Group at Oxford, VGGNet emphasized deeper architectures with smaller convolutional filters (3x3), achieving remarkable accuracy.

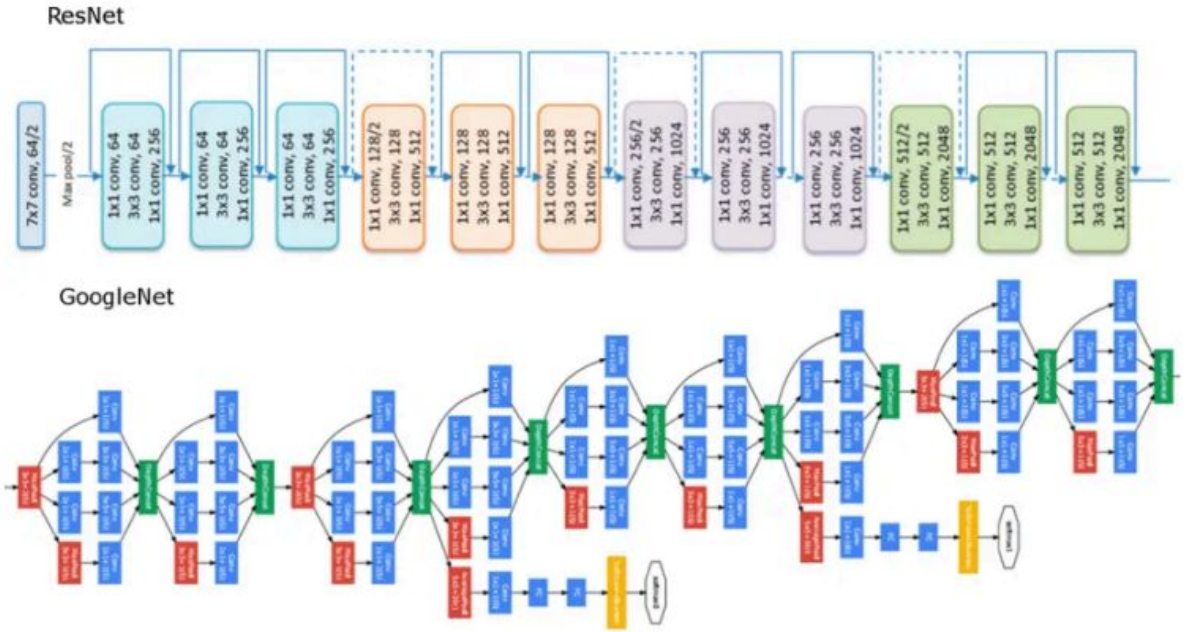
GoogLeNet/Inception (2014): Introduced inception modules that allowed the network to capture multi-scale features efficiently.

ResNet (2015): Residual Networks introduced skip connections, enabling the training of very deep networks while mitigating the vanishing gradient problem.

5.5 Subsequent Architectures



VGG-16 architecture



ResNet and GoogeNet Architectures

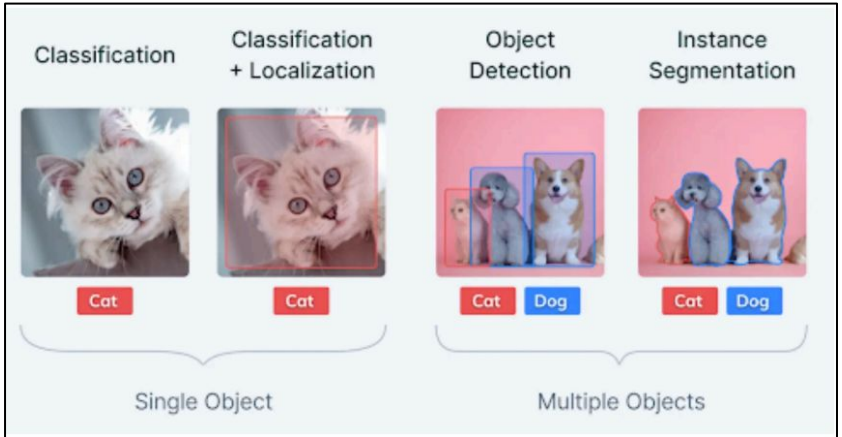
5.6 Applications of CNNs

The advancements in CNNs have revolutionized various fields:

Computer Vision: CNNs have become the backbone of modern computer vision, enabling breakthroughs in image classification, object detection, and semantic segmentation.

Medical Imaging: CNNs are utilized for tasks such as disease diagnosis, tumor detection, and image-guided surgery, significantly improving diagnostic accuracy.

Autonomous Vehicles: CNNs are integral to the perception systems of self-driving cars, allowing them to interpret and respond to their surroundings.



Computer Vision Applications of CNNs.

The journey of CNNs from their inception to their current status as a cornerstone of deep learning illustrates their profound impact on AI.

The success of CNNs has also paved the way for further advancements in deep learning and has inspired the development of other specialized neural network architectures, such as RNNs and Transformers.

The theoretical foundations and practical innovations in CNNs have contributed significantly to the widespread adoption and success of deep learning techniques across various domains.

Hinton, Bengio, LeCun

ACM Awards - Associati…

Geoffrey E Hinton

W Wikipédia

Yoshua Bengio – …

Yann LeCun

LeCun in 2024

FIM